

Fit Time — SDSS

Parametric

Nonparametric

Foundation

n	Parametric												Nonparametric							Foundation			
	LinearGauss-Homo	LinearGauss-Hetero	Student-t	LogNormal-Homo	LogNormal-Hetero	Gamma-GLM	LinGauss-Homo-Ridge	LinGauss-Hetero-Ridge	Student-t-Ridge	LogNormal-Homo-Ridge	LogNormal-Hetero-Ridge	Gamma-GLM-Ridge	FlexCode-RF	FlexZBoost	BART-Homo	BART-Hetero	Flow-Spline	Quantile-Tree	CatMLP	MDN	TabPFN 2.5	RealTabPFN 2.5	TabICL Quantiles
500	<0.01 (<0.01)	0.06s (<0.01)	<0.01 (<0.01)	0.02s (<0.01)	0.04s (<0.01)	1.1s (1.0s)	0.01s (<0.01)	0.06s (<0.01)	0.02s (<0.01)	0.02s (<0.01)	0.03s (<0.01)	0.24s (0.22s)	1.6s (0.51s)	12.8s (8.1s)	4.1s (0.28s)	25.1s (2.2s)	26.3s (2.7s)	9.7s (4.5s)	2.3s (0.39s)	8.0s (3.9s)	0.65s (0.26s)	0.66s (0.27s)	3.7s (2.3s)
1K	<0.01 (<0.01)	0.07s (<0.01)	0.01s (<0.01)	0.03s (<0.01)	0.05s (<0.01)	0.05s (<0.01)	0.01s (<0.01)	0.08s (<0.01)	0.02s (<0.01)	0.04s (<0.01)	0.05s (<0.01)	0.03s (<0.01)	1.7s (0.01s)	4.8s (0.07s)	5.3s (0.37s)	30.4s (2.2s)	41.0s (2.1s)	6.0s (1.6s)	3.8s (0.67s)	6.4s (0.63s)	0.42s (<0.01)	0.42s (<0.01)	2.4s (0.48s)
10K	0.02s (<0.01)	0.18s (<0.01)	0.06s (<0.01)	0.26s (<0.01)	0.30s (<0.01)	0.24s (<0.01)	0.05s (<0.01)	0.19s (0.03s)	0.09s (<0.01)	0.29s (<0.01)	0.30s (<0.01)	0.20s (<0.01)	11.3s (0.40s)	12.3s (1.1s)	32.3s (1.7s)	2.3m (9.7s)	4.4m (19.2s)	26.1m (25.8m)	31.9s (2.7s)	32.7s (2.5s)	2.7s (0.03s)	2.8s (0.04s)	37.8s (2.8s)
50K	0.09s (<0.01)	0.61s (0.02s)	0.29s (<0.01)	1.3s (0.02s)	1.4s (0.01s)	1.0s (0.02s)	0.22s (<0.01)	0.79s (0.04s)	0.42s (<0.01)	1.4s (0.02s)	1.4s (<0.01)	0.94s (0.01s)	1.1m (1.3s)	36.2s (2.6s)	2.8m (8.5s)	12.0m (42.8s)	11.9m (30.0s)	1.9h (40.9m)	3.8m (27.4s)	1.9m (12.5s)	39.9s (3.3s)	41.1s (3.7s)	—
100K	0.16s (<0.01)	1.3s (0.06s)	0.56s (0.02s)	2.5s (0.04s)	2.8s (<0.01)	1.9s (0.05s)	0.45s (0.02s)	1.2s (0.05s)	0.83s (0.03s)	2.7s (0.05s)	2.9s (0.02s)	1.8s (0.03s)	2.3m (1.4s)	1.2m (1.9s)	6.0m (16.9s)	25.1m (1.3m)	17.5m (22.2s)	1.9h (1.0h)	10.9m (47.6s)	3.7m (16.6s)	—	—	—
250K	0.53s (0.12s)	21.4s (1.8s)	1.6s (0.22s)	6.2s (0.17s)	13.1s (0.53s)	10.9s (0.86s)	1.5s (0.10s)	21.0s (0.77s)	2.8s (0.24s)	7.6s (0.31s)	14.8s (0.83s)	6.4s (0.33s)	7.3m (7.7s)	5.2m (0.77s)	17.2m (17.3s)	—	20.8m (23.9s)	—	35.6m (3.5m)	11.6m (45.1s)	—	—	—
500K	1.5s (0.67s)	26.4s (1.8s)	3.9s (1.1s)	12.3s (0.32s)	20.8s (0.62s)	16.2s (0.59s)	3.0s (0.19s)	29.2s (2.1s)	6.3s (1.0s)	14.5s (0.45s)	21.2s (0.59s)	10.9s (0.31s)	18.6m (21.4s)	15.6m (8.8s)	39.7m (2.4m)	—	30.3m (1.3m)	—	1.4h (7.0m)	25.0m (1.5m)	—	—	—